Warpage Modeling of Ultra-Thin Packages Based on Chemical Shrinkage and Cure-Dependent Viscoelasticity of Molded Underfill

Po-Yao Lin and Sanboh Lee

***Abstract*—Molded underfill (MUF) is an essential component in ultra-thin flip chip packages to ensure their long-term reliability and mechanical integrity. However, the warpage evolution of package during the curing process of MUF has a critical effect on both the SMT yield and package reliability. Therefore, a thorough understanding of the MUF physical behavior under curing is essential. The present study therefore successfully establishes a novel process modeling approach based on finite element method to predict the final warpage of an ultra-thin flip chip scale package (fcCSP) in accordance with the chemical shrinkage and cure-dependent viscoelastic behavior of MUF. In describing the cure-dependent behavior of the MUF, the chemical shrinkage, curing kinetics and time-domain viscoelasticity are characterized by the pressure-volume-temperature (PVT) method, differential scanning calorimetry (DSC) and dynamic mechanical analysis (DMA), respectively. The predicted package warpage results are shown to be in good agreement with the experimental thermal Shadow Moiré data. It is additionally shown that the accuracy of the warpage predictions is seriously degraded if the mechanical properties of MUF is modeled using a simple temperature-dependent elastic assumption.**

***Index Terms*—Ultra-thin package, chemical shrinkage, viscoelasticity, finite element method.**

I. Introduction

CONSUMER demand for mobile devices with small form factors and a high performance has driven significant advances in the thinner and 3D IC package integration fields in recent years. Ultra-thin flip chip scale packages (fcCSP) are essential in realizing modern handheld and tablet products, for which device portability and mechanical robustness are vital concerns. As shown in Fig. 1(a), such packages are typically encapsulated using thermosetting polymers, such as epoxy molding compound (EMC) and underfill which offer excellent mechanical and electrical properties to protect the silicon die from damage even under harsh environments and rough handling. In recent year, a new material as we called molded underfill (MUF) to replace both of EMC and underfill is widely used in industry due to its superior reliability and processability. Compared to EMC, MUF has a finer filler size and a superior mechanical strength. As a result, it not only fills the C4 bump area more readily as shown in Fig. 1(b) instead of underfill shown in Fig. 1(a), thereby improving the bump joint reliability, but also provides a better protection of the silicon die. However, the warpage effect induced in the MUF curing process has a serious detrimental effect on the SMT yield and package reliability; particularly for ultra-thin flip chip packages. Accordingly, a thorough understanding of the cure-dependent constitutive behavior of MUF is essential to predict the package warpage induced under typical curing conditions.

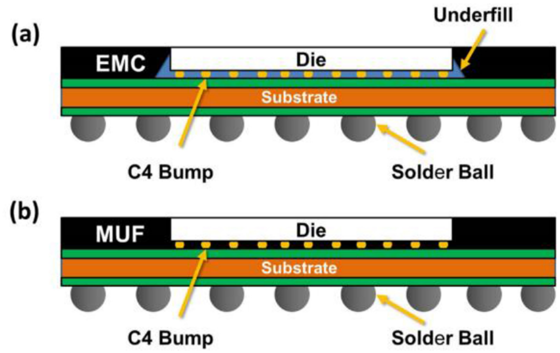

Fig. 1. Schematics of ultra-thin fcCSPs (a) with EMC and underfill (b) with MUF.

The literature contains many studies on the use of viscoelastic-based methods to investigate the stress relaxation behavior of EMC [1], [2], and further the use of finite element modeling method to predict the warpage evolution of electronic packages during the molding process [3]–[6]. But there is almost no relevant study of viscoelastic modeling on MUF until now. Chae *et al.* [1] performed a series of stress relaxation experiments to characterize the viscoelastic behavior of fully-cured EMC. A simple thermo-rheological model was used to deduce the master curve of the EMC relaxation modulus based on a time-temperature equivalence assumption. A prony series expansion was then employed to model the EMC relaxation behavior. Simon *et al.* [2] constructed a model incorporating time-temperature superposition and time-cross-link density superposition principles to predict the development of the viscoelastic properties of the epoxy resin during isothermal

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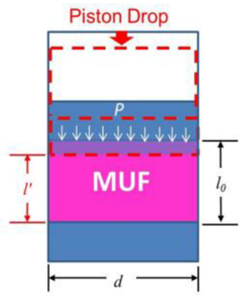

Fig. 2. Schematic illustration of P-V-T testing setup.

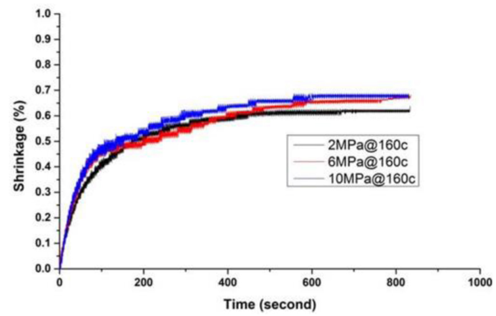

Fig. 3. Experimental chemical shrinkage curves for various pressures and temperature of 160°C.

cure from gelation to after vitrification. The model parameters are obtained from viscoelastic measurements for the fully cured resin and a recursive model for the evolution of the rubbery modulus of the network. Srikanth [3] studied the warpage of epoxy molded packages using viscoelastic based model of fully cured EMC. By using finite element modeling, the effect of compound thickness ratio of the top to bottom side of the package was optimized to reduce the package warpage. Yang *et al.* [4], Chiu *et al.* [5], and Gung *et al.* [6] investigated the cure-dependent viscoelasticity of EMC in the frequency domain by DSC and DMA, then to use finite element modeling method to predict the process-induced warpage of fcCSP packages based on the cure-dependent constitutive model of EMC.

However, the aforementioned literatures almost focus on EMC and no relevant study of viscoelastic modeling on MUF. Besides, the cure-dependent viscoelasticity in time-domain has not been investigated in encapsulated thermosetting polymers yet as well.

Therefore, the present study commences by characterizing the chemical shrinkage and cure-dependent viscoelastic behavior of MUF in the time domain. A novel process modeling approach based on the finite element method incorporated with chemical shrinkage and cure-dependent viscoelastic behavior of MUF is successfully established and then used to predict the curing-induced warpage of ultra-thin fcCSP packages under typical curing conditions of MUF. It is shown that the prediction results are in good qualitative agreement with the experimental thermal Shadow Moiré data.

II. Characterization of Cure Depemdent Constitutive Behavior

The present analysis considers a biphenyl resin-based MUF compound with a filler content of 84 wt.%. The maximum filler size is around 20μm, which is much smaller than that of most common EMC materials (∼ 50μm). The MUF is applied in using transfer molding method consisting of an in-molding process followed by a post molding cure (PMC) operation to ensure a complete polymerization conversion to the fully-cured state. To properly understand the cure-dependent constitutive behavior of the MUF, it is first necessary to characterize its chemical shrinkage and viscoelastic behavior, as described in the following subsections.

A. Chemical Shrinkage

During the molding process, the volume of the MUF shrinks under the effects of a cross-linking reaction. This volume shrinkage can have a significant effect on the package warpage and must therefore be taken into account in any finite element modeling of the MUF molding process. The chemical shrinkage effect can be characterized by means of pressure-volume-temperature (P-V-T) tests performed at different temperatures and pressures [7]. Briefly, an MUF sample is loaded into a P-V-T test system (see Fig. 2) and heated at a known temperature (e.g., 160°C) for a certain period of time (e.g., 1100 sec) under a given pressure P to mimic the chemical reaction which takes place in the in-molding curing process. The chemical shrinkage is then calculated from the measured piston drop in accordance with

$$\textit{Chemical Shrinkage} = \frac{\left|V' - V_0\right|}{V_0} \times 100\% = \frac{\left|l' - l_0\right|}{l_0} \times 100\% \quad (1)$$

where V_0 and l_0 are the volume and height of the original MUF pallet; V' and l' are the reduced volume and height of the MUF pallet after a certain testing time, respectively; and $|l' - l_0|$ is the corresponding drop in the piston height.

Fig. 3 shows the typical results obtained from a P-V-T test performed at a temperature of 160°C. For each of the considered pressures, the chemical shrinkage rate increases dramatically at the beginning of the simulated curing process and then stabilizes after approximately 600 sec. In other words, the cross-linking reaction takes the MUF from a viscous flow state to a rubbery state. It is observed from Fig. 3 that the chemical shrinkage increases with an increasing pressure. For example, as shown in Table I, the final shrinkage rate (i.e., the MUF has a fully-cured state) has values of 0.62%, 0.65% and 0.68% for curing pressures of 2 MPa, 6 MPa and 10 MPa, respectively.

B. Curing Kinetics

In general, the level of curing reaction for a polymer processed under a certain temperature history can be characterized by the degree of cure (DOC), which is defined

$$\alpha = \frac{H(t)}{H_u} \quad (2)$$

TABLE I
CHEMICAL SHRINKAGE WITH VARIOUS PRESSURES AT ALMOST FULLY CURED LEVEL

Pressure (P)	Chemical Shrinkage
2MPa	0,62%
6MPa	0.65%
10MPa	0.68%

TABLE II
KAMAL AUTOCATALYTIC MODEL CONSTANTS

Constant	Fitting Results
A_1	1.0147E+02 /sec
A_2	1.9051E+06 /sec
$\Delta E_1/R$	6.5571E+03K
$\Delta E_2/R$	8.0093E+03K
m	3.0746E-01
n	1.0089E+00

where $H(t)$ is the heat released by the cross-linking reaction over time t and H_u is the ultimate heat of reaction.

To measure the heat of reaction of the MUF, and develop the associated curing kinetics model, DSC experiments were performed using a TA Instruments Q200 system to measure the heat flow of the uncured MUF under a constant heating rate. The heat flow data were then integrated over time to determine the ultimate heat of reaction and DOC of the MUF under a certain temperature history. The curing kinetics of the MUF were then described using the Kamal autocatalytic model [8], i.e.,

$$\frac{d\alpha}{dt} = \left(K_1 + K_2\alpha^m\right) \times (1-\alpha)^n \tag{3}$$

where α is the DOC, m and n are the reaction orders, and K_1 and K_2 are reaction rate constants. Both constants have an Arrhenius-type temperature dependence, i.e.,

$$K_i = A_i e^{\left(\frac{-\Delta E_i}{RT}\right)} i = 1, 2 \tag{4}$$

where A_i is a constant, ΔE_i is the activation energy of the MUF material, R is the universal gas constant and T is the absolute temperature. In the present study, the Kamal autocatalytic constants were determined by fitting the DSC experimental data obtained under constant heating rates of 10°C/min, 20°C/min, 40°C/min and 60°C/min, respectively. The results are shown in Table II.

The glass transition temperature (Tg) associated with different DOCs was also characterized via DSC experiments in accordance with the following modified empirical DeBenedetto equation [2]:

$$\frac{T_g - T_{g0}}{T_{g\infty} - T_{g0}} = \frac{\lambda\alpha}{1-(1-\lambda)\alpha} \tag{5}$$

where T_{g0} and $T_{g\infty}$ are the glass transition temperatures of the uncured and fully-cured MUF, respectively, and λ is a constant. Fig. 4 compares the experimental results obtained for the variation of the glass transition temperature with the DOC with the results obtained via model fitting. Note that in implementing the DeBenedetto model, T_{g0} and $T_{g\infty}$ were assigned values of 303 K (30°C) and 393 K (120°C), respectively. A fitting process revealed that the constant λ in the DeBenedetto model had a value of approximately 0.2294.

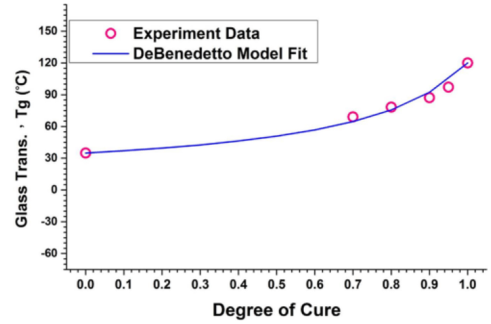

Fig. 4. Comparison of experimental results and DeBendetto predictions for variation of Tg with DOC.

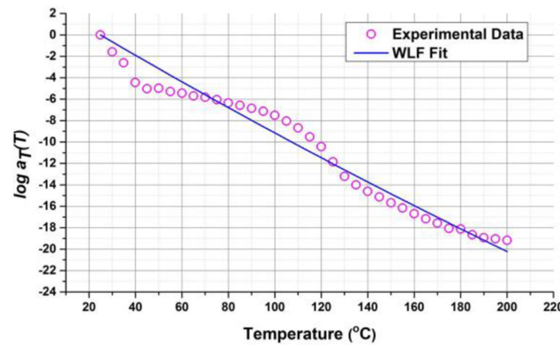

Fig. 5. Experimental fitting of time-temperature shift factor using WLF model.

C. Cure-Dependent Viscoelasticity

The viscoelastic behavior of the MUF was investigated by stress relaxation experiments performed in the time domain using the DMA technique. The stress relaxation storage modulus of the fully-cured MUF was obtained over a period of 600 sec at various temperatures in the range of 25∼200°C. A single master curve was then constructed by setting the reference temperature as 25°C and shifting the relaxation modulus vs. time curves obtained at various temperatures in the horizontal direction relative to the reference curve. In other words, the master curve was constructed using a time-temperature superposition (TTS) assumption based on the time-temperature shift factors obtained from the following Williams–Landel–Ferry equation:

$$\log a_T(T) = \frac{-C_1(T-T_r)}{C_2 + (T-T_r)} \tag{6}$$

where T_r is the reference temperature (298 K); and C_1 and C_2 are model constants obtained via experimental fitting. For the present MUF, the model constants were found to have values of 220 K and 1731 K, respectively (see Fig. 5).

The MUF master curve can be approximated using a generalized Maxwell model referred to as the prony series, which considers the joint effects of time, temperature and the DOC,

TABLE III
GENERALIZED MAXWELL MODEL CONSTANTS

E_0 = 15 GPa					
i	ω_i	τ_i *(sec)*	*i*	ω_i	τ_i *(sec)*
1	6.54E-11	1.0E-01	8	5.94E-02	1.0E+17
2	3.32E-02	1.0E+04	9	2.14E-02	1.0E+19
3	7.15E-02	1.0E+07	10	2.28E-03	1.0E+20
4	1.45E-01	1.0E+09	11	8.46E-03	1.0E+21
5	2.96E-01	1.0E+11	12	6.77E-07	1.0E+22
6	2.15E-01	1.0E+13	13	5.93E-03	1.0E+23
7	1.16E-01	1.0E+15	14	1.20E-02	1.0E+26

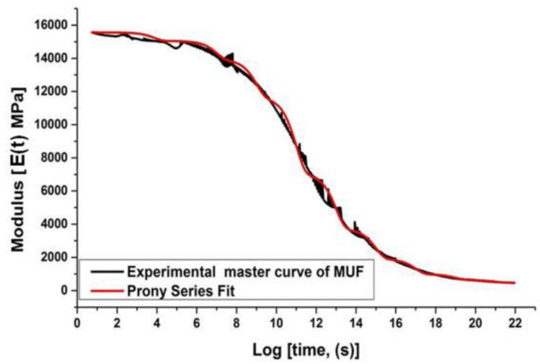

Fig. 6. Experimental MUF master curve versus generalized maxwell model predictions.

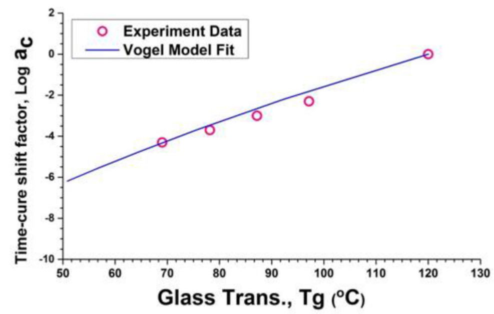

Fig. 7. Comparison of experimental and prediction results for variation of time-cure shift factor with Tg.

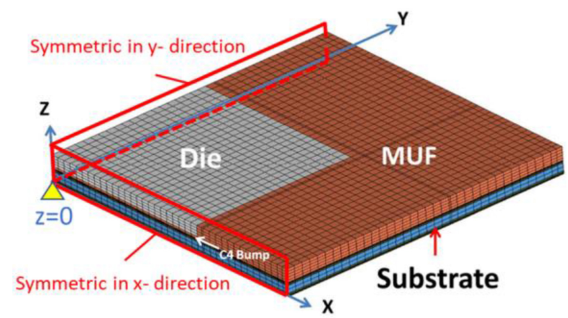

Fig. 8. Quarter FE model of ultra-thin fcCSP package.

and has the form [5]:

$$E(t, T, \alpha) = E(\xi, T_T, \alpha) = E_0(\alpha)\left[\omega_\infty(\alpha) + \sum_{i=1}^{N} \omega_i e^{\left(-\frac{\xi}{\tau_i(\alpha)}\right)}\right] \tag{7}$$

where ξ is the reduced time or pseudo-time; T_T is the reference temperature for the time–temperature correspondence, E_0 is the glassy modulus of the MUF, *N* is the number of Maxwell elements, τ_i is the relaxation time, ω_i is a weighting factor for the ith Maxwell element, and $\omega_8 + \Sigma\omega_i = 1$. Note that the relationship between the real and reduced time is given by

$$\xi = \int_0^t \frac{T(\eta)}{a_T} d\eta \tag{8}$$

where a_T is the time-temperature shift factor. In practice, the master curve of the fully-cured MUF can be well-fitted with the Maxwell model (Eq. (7)) when $\alpha = 1$. Table III shows the fitted parameters obtained for the prony series when comparing the experimental master curve for the present MUF model with the prediction results obtained using the generalized Maxwell model (see Fig. 6).

The cure-dependent master curve can be modeled using an approach similar to that described above for the time-temperature dependence. The present study considers conversion values in the range of α = 0.7∼1.0. And an assumption was made that the generalized Maxwell model was insensitive to the DOC and only the relaxation time varied as a function of α. The dependence of the relaxation time on the DOC was modeled as [5]

$$\tau_i(\alpha) = a_c(\alpha)\tau_i^* \tag{9}$$

where τ_i* is the relaxation time for Maxwell element *i* when $\alpha = 1$ and a_c is the time-cure shift factor described by the following Vogel equation [2]:

$$a_c = e^{\left[-\frac{\Delta E}{R}\left(\frac{1}{T_g(\alpha)} - \frac{1}{T_{g\infty}}\right)\right]} \tag{10}$$

The value of the time-cure shift factor a_c for different DOCs was obtained by constructing the master curves for different DOC samples and then shifting these curves along the time axis until they overlapped the fully-cured master curve. The relationship between T_g and the DOC was derived from Eq. (5). The variation of the time-cure shift factor a_c with T_g was then obtained with the form shown in Fig. 7, in which the constant $\Delta E/R$ in the Vogel model (Eq. (10)) was found from experimental fitting to have a value of 2.62E+4 K.

III. WARPAGE PREDICTIONS BY FINITE ELEMENT MODELING

A. Package Model Establishment

A three-dimensional FE model of a typical ultra-thin flip chip scale package (fcCSP) was constructed in ANSYS [9], as shown in Fig. 8.

The model comprised three components, namely the silicon die, MUF and the substrate. The overall package had dimensions of 16 mm (L) x 16 mm (W) x 0.68 mm (H), while the die measured 10 mm (L) x 10 mm (L) x 0.25 mm. The height of the C4 bumps between the die and the substrate was set as 0.06 mm and the substrate thickness was given as 0.37 mm. The model was meshed using 8-node solid elements. Due to the symmetric nature of the package, only one quarter of the package was modeled for analysis purposes. The symmetric conditions are set at x-direction and y-direction,

TABLE IV
MECHANICAL PROPERTIES IN FE MODEL

Material	E	ν	CTE
Die	169.54	0.278	3.5
Substrate	18.62	0.42	16
MUF (linear elastic)	16.2@25°C 13.2@100°C 0.75@150°C 0.16@250°C	0.26	20@25°C 35@100°C 87@150°C 100@250°C

E is the elastic modulus with units of GPa, ν is the Poisson ratio, and CTE is the coefficient of thermal expansion with units of ppm/K.

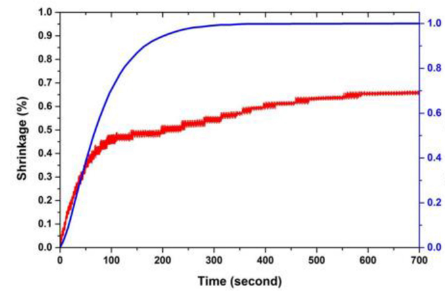

Fig. 9. Experimental chemical shrinkage under pressure of 6 MPa and predicted DOC at 160°C.

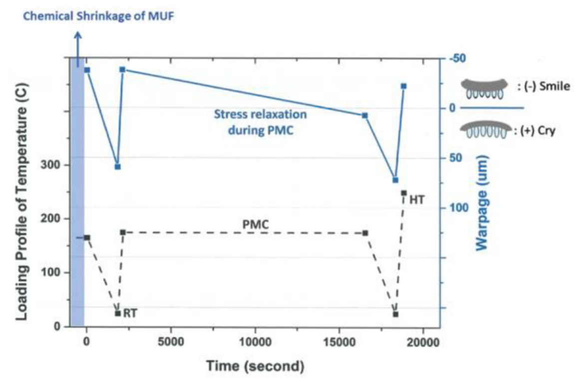

Fig. 10. Warpage predictions for the temperature profile of the in-molding process, PMC process, and Shadow Moiré warpage measurement process.

respectively. The displacement is constrained (z = 0) in the center of the bottom of package to avoid the movement along z-direction. The layer of C4 bump is assumed by MUF because the bump density is 9% of the volume in the layer of C4 bump. Therefore, it is almost filled with MUF and we can assume MUF for it reasonably in model.

The cure-dependent viscoelastic properties of the MUF were specified using the material models established in the previous section. In addition, the DOC was assigned an initial value of $\alpha = 0.7$ since generally the mechanical stiffness of encapsulation materials starts to develop only at the gel point of approximately $\alpha = 0.7$. The mechanical properties of the die and substrate were assigned the values shown in Table IV. For comparison purposes, the MUF was modeled using both the cure-dependent viscoelastic properties derived in the previous section and the linear elastic properties obtained via DMA at temperatures in the range of 25 ∼ 250°C is also shown in Table IV.

B. Process Modeling and Warpage Validation

The loading condition comprised two stages, namely linear chemical shrinkage (determined by the initial strain applied in the first loading step of the model) and thermal loading (determined by the temperature profile imposed in the subsequent in-molding process and PMC process). The stress-free temperature was set as 160°C of the in-molding process temperature.

1) Initial Strain: As stated above, the linear chemical shrinkage effect was assumed to impact the package warpage behavior only after $\alpha = 0.7$. The volume chemical shrinkage effect described in Section II-A can be expressed in terms of the linear shrinkage in the x-, y- and z-directions as follows:

$$V_c = \frac{|V' - V_0|}{V_0} = \frac{|(L_0 - \Delta L)^3 - L_0^3|}{L_0^3} = \left|\left(1 - \frac{\Delta L}{L_0}\right)^3 - 1\right| \tag{11}$$

where *V'* is the shrunk volume after the molding process, ΔL is the displacement in each direction, and $\Delta L/L_0$ is the isotropic cure-induced strain in the corresponding direction (i.e., the linear shrinkage (ε_c)). Assuming that the linear shrinkage and pressure variation in the volume are sufficiently small and isothermal temperature conditions prevail for a specific packing process, the MUF volume shrinkage can be simplified to the following strain form:

$$V_c = \left|\left(1 - \frac{\Delta L}{L_0}\right)^3 - 1\right| \cong \left|1 - 3\left(\frac{\Delta L}{L_0}\right) - 1\right| = 3\varepsilon_c \tag{12}$$

where V_c is the volume chemical shrinkage. In other words, the linear chemical shrinkage is equal to one-third of the volume chemical shrinkage.

Fig. 9 shows the change in the volume chemical shrinkage over time under a pressure of 6 MPa and a temperature of 160 °C. The corresponding change in the DOC of the MUF (as evaluated using the curing kinetics model with the fitting constants listed in Table II) is also shown for reference purposes. As shown, the MUF volume change induced by polymerization (i.e., $\alpha = 0.7$) commences after approximately 100 sec. The effective shrinkage assigned to the initial strain in the FE model can be taken as the difference in the chemical shrinkage between $\alpha = 0.7$ and $\alpha \approx 1$ (the fully-cured state), respectively. In other words, based on the results presented in Fig. 9, the initial strain value in the present study was set as (0.65-0.45)% /3 = 0.067 %.

2) Thermal Profile: In simulating the warpage induced in the assembly process, the temperature profile for the in-molding process, PMC process, and Shadow Moiré warpage measurement process was assigned as follows: (i) isothermal curing at 160°C (MUF curing from $\alpha = 0.7$ to almost fully cured condition) for 50 sec to simulate the in-molding process; (ii) cooling from the in-molding temperature of 160°C to 25°C in 30 min; (iii) heating from 25°C to the PMC temperature 175°C in 300 sec; (iv) curing at 175°C for 4 hr to simulate

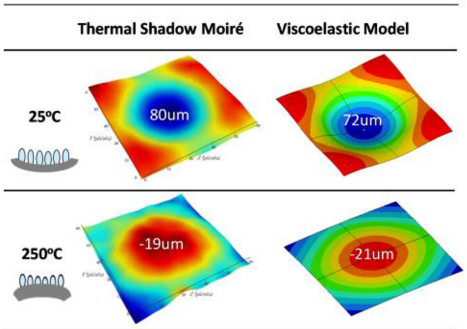

Fig. 11. Experimental and simulated warpage contours at 25°C and 250°C.

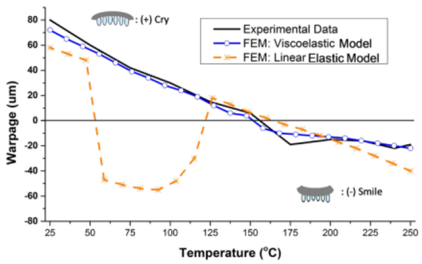

Fig. 12. Warpage evolution of ultra-thin fcCSP from 25°C to 250°C.

the PMC process; (v) cooling from 175°C to 25°C in 30 min; and (vi) heating from 25°C to 250°C in 480 sec to simulate thermal Shadow Moiré warpage measurement process.

3) Warpage Validation: The warpage predictions for the temperature profile from (i) to (vi) is shown in Fig. 10. It's noticed that the warpage level is changed from −37.9um with smile shape to 7.4um with cry shape during PMC process due to stress relaxation at 175C with 4 hrs. This is what linear elastic model can not predict at constant temperature to result in the deviation of warpage prediction for the following process. Fig. 11 compares the experimental thermal Shadow Moiré contours of the fcCSP at 25°C and 250°C, respectively, with the results obtained from the proposed viscoelastic model. It is seen that a good quantitative agreement exists between the two sets of results at both temperatures. Fig. 12 compares the predicted package warpage at various temperatures with the experimental measurements. Note that the predictions are obtained using two different models for the mechanical properties of the MUF, namely the viscoelasticity model developed in Section II and the simple linear temperature-dependent elasticity model commonly adopted in industry (shown in Table IV). The prediction results obtained using the proposed viscoelastic model are in good agreement with the experimental measurements over the entire temperature range. However, the results obtained using the conventional temperature-dependent elasticity model deviate significantly from the experimental results; particularly at temperatures in the range of around 50 ∼ 125°C. We take the warpage comparison at 75°C to interpret the physical insight. And the prediction at 25°C is also compared as well. As shown in Fig. 13, for 25°C, it can be seen that the warpage level of viscoelastic model is 71um higher than 59um of elastic model. And the viscoelastic model

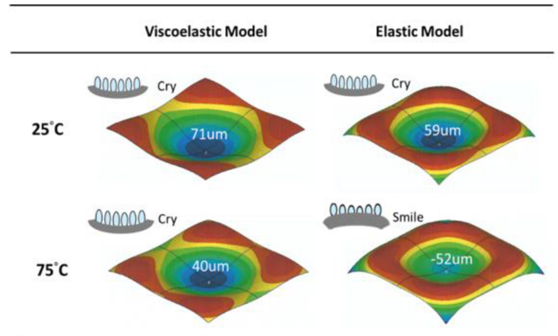

Fig. 13. Warpage comparison of viscoelastic model vs. elastic model at 25°C and 75°C.

shows the corners on the package is slightly cry shape but elastic model shows the smile shape which is induced by the high modulus and thermal shrinkage of MUF due to no stress relaxation behavior in MUF. When temperature is increased to 75C, the warpage is decreased to 40um in viscoelastic model by the effect of thermal mismatch minimized between die/MUF and substrate. On the contrary to elastic model, if we don't consider stress relaxation effect in MUF, the interaction of thermal mismatch between MUF and substrate will be stronger than die and substrate to result in higher smile shape on the corner and lower cry shape on the center of the package. Therefore, the warpage level becomes −52um with smile shape. That's the reason why the elastic model gets poor agreement with experimental data.

IV. Conclusion

This study has performed a detailed characterization of the cure-dependent constitutive behavior of MUF including chemical shrinkage and cure-dependent viscoelasticity. The material model of cure-dependent viscoelasticity in time domain has been successfully developed by constructing master curve described by prony series and cure-temperature shift factors based on characterization results. Then the effective shrinkage and material model of cure-dependent viscoelasticity have been employed in FE model for predicting the curing-induced warpage of ultra-thin fcCSPs under typical MUF curing process conditions. It has been shown that the warpage of ultra-thin package evolved during reflow process is highly sensitive to MUF due to its high volume occupied in package. More importantly, the prediction results obtained using the proposed process modeling approach in FE model are in good agreement with the experimental Shadow Moiré data. By contrast, the predictions obtained using the conventional linear elastic assumption for the MUF behavior deviate significantly from the experimental measurements. Consequently, the importance of adopting a cure-dependent constitutive model including chemical shrinkage and cure-dependent viscoelasticity for predicting the warpage of ultra-thin packages such as fcCSPs or fan-out package is confirmed.

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Conflicts of Interest

The authors declare that there are no conflicts of interest.

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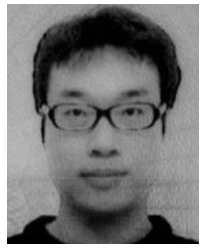

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